

Vansh Saini — Portfolio Website

Complete Technical Documentation & Performance Analysis
Prepared: July 2026 · B.Tech CSE (AI/ML) · Quantum Global Campus, Roorkee

1. Overview

Detail	Information
Owner	Vansh Saini
Role	Software Engineer · ML Developer
Email	vanshsaini8457@gmail.com
Phone	+91 7895745198
Location	Roorkee, Uttarakhand, India
GitHub	github.com/vanshsaini
LinkedIn	linkedin.com/in/vansh-saini-a0bb193382
Project Type	Frontend-only React SPA (no backend / no database in site itself)

2. Website Sections

Section	Component	Purpose
Loading Screen	Loading.tsx	Asset preload, progress bar, intro unlock animation
Landing / Hero	Landing.tsx	Name, role titles, animated text swap
3D Character	Character/Scene.tsx	Interactive rigged 3D model with head tracking
About	About.tsx	Bio, education, skills summary
What I Do	WhatIDo.tsx	Full Stack & AI/ML skill cards
Career	Career.tsx	Timeline of internships & hackathons
Work	Work.tsx	Project carousel with live links
Tech Stack	TechStack.tsx	3D physics scene with tech-logo spheres (desktop only)
Contact	Contact.tsx	Contact form (mailto), social links, footer
Navbar	Navbar.tsx	Navigation + ScrollSmoother wrapper
Social Icons	SocialIcons.tsx	GitHub, LinkedIn, certs, resume download
Custom Cursor	Cursor.tsx	Animated GSAP cursor with hover states

3. Tech Stack

Core Frontend

Technology	Version	Used For
React	18.3	UI components, state, lazy loading
TypeScript	5.5	Type-safe development
Vite	5.4	Dev server & production bundling
HTML5 / CSS3	—	Structure & all component styling

Animation & Scroll

Library	Used For
GSAP	Core animation engine — tweens, timelines, easing
ScrollTrigger	Scroll-linked section & character animations
ScrollSmoother	Smooth scroll (smooth: 1.7, effects: true)
SplitText	Char/word/line split for staggered text reveals
react-fast-marquee	Loading screen scrolling text

3D & WebGL

Library	Used For
Three.js	Vanilla WebGL renderer for hero 3D character
three-stdlib	GLTFLoader, DRACOLoader, RGBELoader
@react-three/fiber	React renderer for Tech Stack 3D scene
@react-three/drei	HDR Environment map
@react-three/rapier	Physics — 30 floating tech spheres
@react-three/postprocessing	N8AO ambient occlusion pass
DRACO	Compressed 3D mesh decoding

4. Programming Languages & Tools

Language / Tool	Where Used
TypeScript	All React components, utils, context
JavaScript	GSAP scripts, Three.js, DRACO decoder
CSS	15 component stylesheets + global CSS
React.js	Core UI framework + showcased skill
Next.js	Showcased in Work projects & What I Do
Node.js	What I Do, CRM project, Tech Stack sphere
Express.js	AI-Powered CRM project (external)
Python	What I Do — ML section (scikit-learn, NumPy, Pandas)
Tailwind CSS	External showcased projects
Chart.js	AI-Powered CRM project
Git / GitHub	Version control + social link
Generative AI / Copilot Studio	Certifications & What I Do

5. Databases

Note: This portfolio website does **NOT** use any database. It is a static frontend SPA. Contact form uses `mailto:`.

Database	Where Used	Purpose
MongoDB	AI-Powered CRM System (showcased project)	Backend data storage for CRM app
MySQL	Tech Stack 3D sphere texture	Skill showcase only

6. Color Palette

Color Name	Hex Code	Usage
 Background	#0a0e17	Main dark navy background
 Accent	#5eead4	Links, highlights, buttons, headings
 Body Text	#eae5ec	Default paragraph text
 Initial BG	#050810	Pre-loading darker background
 Cyan	#22d3ee	3D point light, landing circles
 Teal	#14b8a6	Landing text, career timeline
 Bright Teal	#2dd4bf	Loading screen accent
 Mint	#e0f2f1	Loading screen background
 Purple	#e6c3ff	Custom cursor
 Shirt Brown	#8B4513	3D character shirt mesh
 Pants Black	#000000	3D character pants mesh
 Screen Glow	#B0F5EA	3D monitor emissive light

Typography

Font: Geist (Google Fonts) — used for all headings and body text.

7. Animations & Motion

Loading Screen

- Progress counter (0→100%) with simulated load intervals
- Marquee: "Software Engineer / ML Developer" scrolling text
- Mouse-follow glow via CSS variables
- Loader ball game CSS animation
- GSAP background fade: #050810 → #0a0e17
- SplitText char reveal with blur + y-offset stagger
- Infinite "Engineer ↔ ML Dev" text swap loop

Smooth Scroll (ScrollSmoother)

Setting	Value	Purpose
smooth	1.7	Interpolated smooth scrolling
speed	1.7	Scroll speed multiplier
effects	true	Parallax-style scroll effects
Starts paused	true	Waits until loading screen completes

Character Scroll Story (GsapScroll.ts)

Timeline	Trigger	Animates
tl1	Landing → About	Character rotation, camera Z, slides left, landing fades
tl2	About → What I Do	Camera fly, monitor lights up, neck bone rotates
tl3	What I Do	Character exits viewport, section parallax
careerTimeline	Career	Timeline grows, info boxes stagger in

Other Animations

- **SplitText:** .para words & .title chars fade up on scroll
- **Custom Cursor:** GSAP-smoothed follow with hover expand/hide
- **Social Icons:** Magnetic icon effect per icon (rAF loop)
- **Work Carousel:** CSS translateX slide with dot navigation
- **Career Dot:** CSS box-shadow pulse glow animation
- **What I Do Cards:** Touch click-to-expand on mobile

8. 3D Character System

File	Purpose
Scene.tsx	WebGL renderer, camera, animation loop, mouse input
character.ts	GLTF load, DRACO decode, shirt/pants colors
lighting.ts	HDR env map, directional + point lights
animationUtils.ts	Mixer: intro, typing, blink, keyboard, eyebrow
mouseUtils.ts	Head bone follows mouse with lerp
decrypt.ts	AES decrypt for encrypted model (character.enc)

Character GLTF Animations

Clip	Purpose
introAnimation	Character intro pose on load
key1, key2, key5, key6	Keyboard key press animations
typing	Hand/arm typing (filtered finger bones)
Blink	Eye blink loop after intro
browup	Eyebrow raise on face hover

3D Lighting

Light	Color	Details
Directional	#5eead4	Main light + 1024px shadow map
Point	#22d3ee	Screen glow tied to monitor emissive
HDR Environment	char_enviorment.hdr	Image-based reflections
Tone Mapping	ACES Filmic	Cinematic color grading

9. Tech Stack 3D Scene (Desktop Only >1024px)

Feature	Detail
Renderer	React Three Fiber Canvas
Physics	Rapier — 30 rigid body spheres
Post-processing	N8AO ambient occlusion (#0f002c)
Interaction	Spheres follow mouse past Work section
Shadows	Spot light 512px shadow map

Tech Logo Spheres

Texture	Technology
react2.webp	React.js
next2.webp	Next.js
node2.webp	Node.js
express.webp	Express.js
mongo.webp	MongoDB
mysql.webp	MySQL
typescript.webp	TypeScript
javascript.webp	JavaScript

10. Projects Showcased

#	Project	Category	Tech Used
1	Royal Spice Restaurant	Business Website	React.js, Tailwind CSS, JavaScript, Vercel
2	AI-Powered CRM System	Full Stack App	JavaScript, Chart.js, Node.js, Express.js, MongoDB
3	Personal Portfolio Website	Web Development	Next.js, React, TypeScript, Tailwind CSS, Vercel
4	Viraasat — Ethnic Wear Store	Hackathon E-Commerce	Next.js, React, TypeScript, Tailwind CSS, Context API

11. Certifications

Certification	Provider
What Is Generative AI?	LinkedIn Learning
Microsoft Copilot Studio	Microsoft
Excel Automation using ChatGPT	Professional Certification
National Financial Literacy	Certification Program

12. Career Experience

Role	Organization	Year
Outreach & Sponsorship Associate	HackIndia	2026
Core Organizer	Q-Hackathon 2026, Quantum University	2026
Full Stack Web Development Intern	Future Interns	2026
Core Organizer	Crazy UX AI Challenge Hackathon	2026
Core Member	CodeX Club, Quantum University	2025

13. Performance — Why Website Lags (Atakti Hai)

Bottom Line: MacBook slow nahi hai — site design bahut ambitious hai. Ek saath cinematic scroll, rigged 3D character, physics scene, aur text animations chalte hain. Frame budget exceed ho jata hai, isliye 60fps nahi milta.

Main Reasons (Priority Order)

Rank	Reason	Impact	Explanation
1	Do WebGL scenes ek saath	Very High	Character (Three.js) + TechStack (R3F) dono GPU par simultaneously render hote hain jab Work section ke paas scroll karo
2	Character hamesha render hota hai	Very High	requestAnimationFrame loop kabhi band nahi hota — screen se bahar ho ya tab hidden ho, phir bhi GPU draw karta hai
3	Full Retina pixel ratio	High	devicePixelRatio 2x = 4x zyada pixels. Detailed model + shadows + antialias = bahut heavy
4	Scroll aur 3D alag loops par	High	GSAP ScrollSmoother ek ticker par, Three.js alag rAF par — sync nahi = scroll pe jerk
5	TechStack physics + N8AO	High	30 Rapier bodies + N8AO post-processing + shadows har frame
6	ScrollSmoother heavy settings	Medium	smooth: 1.7 + effects: true = har frame extra DOM transforms
7	Multiple rAF loops	Medium	Cursor loop + 3 SocialIcons loops + Character loop + R3F loop = 6+ parallel loops
8	GPU-heavy CSS	Medium	blur() text animation, box-shadow glow pulse, mix-blend-mode cursor

Safe Fixes (Without Breaking Animations)

- Character render sirf tab jab screen par visible ho (Intersection Observer)
- TechStack canvas sirf Work section ke paas load karo
- Pixel ratio cap: `Math.min(devicePixelRatio, 2)`
- Three.js render ko GSAP ticker se sync karo (alag rAF ki jagah)
- Tab hidden hone par render pause karo

14. Getting Started

Command	Description
npm install	Install all dependencies
npm run dev	Start Vite dev server (network accessible)
npm run build	TypeScript compile + production build
npm run preview	Preview production build locally
npm run lint	Run ESLint

15. Responsive Breakpoints

Breakpoint	Behavior
> 1024px (Desktop)	Full 3D character outside scroll + TechStack visible
≤ 1024px (Tablet/Mobile)	Character inside Landing, TechStack hidden
≤ 900px	SplitText animations disabled
Touch devices	What I Do cards use click-to-expand